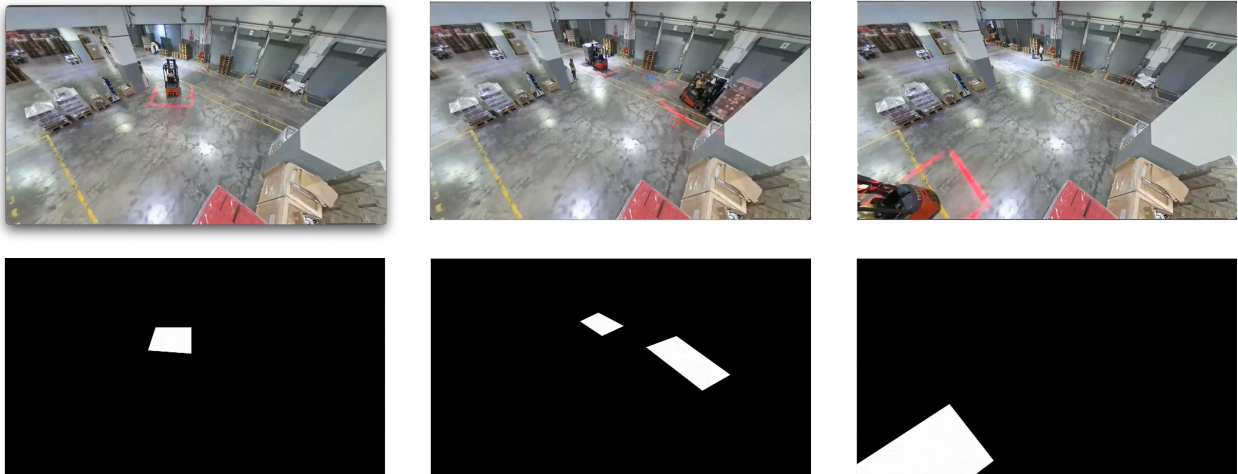


Technical Challenge: Detect clean red laser boundary around forklifts

Welcome to the second round of interviews for Zapdos Labs for the role of Founding Computer Vision Engineer. For this round, we want to test your technical skills in the Computer Vision domain, which helps us evaluate your coding & problem solving skills. If you pass this round, you would have a short final interview with the team to see how you could fit in & for us to answer any culture-related questions you might have.

This challenge is inspired by one of the early problems we faced, which we think is conceptually simple to explain, but has a lot of nuances to consider when implementing.



Given a dataset with a limited number of video samples, how would you design a Computer Vision algorithm to mask the red laser boundary around the forklifts? You are allowed to use any models or image processing algorithms. Manual parameter tuning is allowed. You would need to implement a functional pipeline that takes in a video, and outputs a binary (black and white) masking video (min 2 FPS).

Example: In the frames from the video above, the top-left frame shows a forklift with red laser lines forming a rectangle around it. In the bottom-left frame, that rectangle is masked cleanly as a white rectangle.

Some of the challenges that you might encounter:

- Noisy output: It's important for your algorithm to smooth out the result and not produce particle dust / speckle
- Perspective warping & angle warping.
- Performance: Running VLM/big models on every frame is expensive. If you need to, run only at the very start (first few frames e.g. for depth estimation). Else, you can run smaller ones like MobileSAM, open vocabulary detection models, or hand designed CV algorithms.

You will find the video samples (main.mp4 files) at <https://github.com/zapdos-labs/technical-interview>. We have included some sample output clips for illustrative purposes; we expect the final algorithm to output cleaner masks rendered as filled rectangles instead of just outlines

We kept some clips from the same factory site that we want to test your algorithm on. After you are done, put your code in a private GitHub repo, add <https://github.com/tri2820> as a viewer, and email me at tri@zapdoslabs.com to signal you are done. After that, please schedule a call at <https://calendar.app.google/pB7kKrUmGtRk2PCt5> so we can walk through your implementation together.

You would have 10 days to work on this, but we designed this exercise so you can complete it in 1 or a few days. The test aims to reveal the candidate's approach to problems we face daily, so we care more about how you did it, does it make sense logically, instead of the immediate perfect output, though that is **much** appreciated.

As a startup, we prioritize speed. Write in any language that is most comfortable to you, and we would love to see if you used any AI coding assistant to help you implement / brainstorm.

For any questions, you can also reach out to the above email address.